

Taxation and Information

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1 Motivation

We seek to understand how information disclosure affects optimal outcomes across two types of individual, high and low-types. We employ components of logic from the Stieglitz model. We first begin by solving a monopoly problem, before applying the results obtained to an information design setting with three different planners: a Rawlsian (who prefers the low-type individual), a Randian (who prefers the high-type individual), and a Utilitarian (who weights both equally). This process is split into three files: `screening_Problem`, `welfare_alpha`, and `secant_tangent`.

2 Screening Problem

2.1 Setup

A monopolist serves a population of consumers with private valuations $\theta \in \{\theta_L, \theta_H\}$, where $\theta_H = 10$ and $\theta_L = 3$. A share $\lambda \in (0, 1)$ of consumers are low-types and $(1 - \lambda)$ are high-types. Consumer utility is quasi-linear:

$$U(q, p; \theta) = \theta q - p$$

and the monopolist faces a convex cost function $c(q) = \frac{1}{2}q^2$, so marginal cost equals q .

2.2 Monopolist's Problem

The monopolist offers a menu $\{(q_H, p_H), (q_L, p_L)\}$ to maximize expected profit:

$$\max_{q_H, q_L, p_H, p_L} (1 - \lambda) \left[p_H - \frac{1}{2}q_H^2 \right] + \lambda \left[p_L - \frac{1}{2}q_L^2 \right] \quad (1)$$

subject to the standard incentive-compatibility and participation constraints:

$$\theta_L q_L - p_L \geq 0 \quad (2)$$

$$p_H = \theta_H q_H - (\theta_H - \theta_L) q_L \quad (3)$$

2.3 Analytical Solution

At the optimum both constraints bind. From (1),

$$p_L = \theta_L q_L \quad (4)$$

Substituting into (2):

$$p_H = \theta_H q_H - (\theta_H - \theta_L) q_L \quad (5)$$

Inserting both into the objective gives

$$\max_{q_H, q_L} (1 - \lambda)[\theta_H q_H - (\theta_H - \theta_L) q_L - \frac{1}{2} q_H^2] + \lambda[\theta_L q_L - \frac{1}{2} q_L^2] \quad (6)$$

From this, we can obtain the first-order conditons:

$$\frac{\partial}{\partial q_H} = (1 - \lambda)(\theta_H - \theta_L) + \lambda(\theta_L - q_L) = 0 \quad (7)$$

The high-type receives its first-best quantity with no downward distortion (*no distortion at the top*). The low-type's quantity is distorted downward to reduce the information rent that must be conceded to the high type; exclusion occurs ($q_L^* = 0$) when λ is sufficiently small, specifically when

$$\lambda < 1 - \frac{\theta_L}{\theta_H} = 0.70$$

2.4 Equilibrium Prices and Payoffs

Optimal prices follow directly from the binding constraints:

$$p_L^* = \theta_L q_L^*, \quad p_H^* = \theta_H q_H^* - (\theta_H - \theta_L) q_L^*$$

Consumer utilities are:

$$U_L = \theta_L q_L^* - p_L^* = 0, \quad U_H = \theta_H q_H^* - p_H^* = (\theta_H - \theta_L) q_L^* \geq 0$$

$U_L = 0$ confirms the low-type's participation constraint binds exactly; $U_H \geq 0$ is the information rent earned by the high-type. Total profit is:

$$\pi(\lambda) = (1 - \lambda)[p_H^* - \frac{1}{2}(q_H^*)^2] + \lambda[p_L^* - \frac{1}{2}(q_L^*)^2]$$

Under the UBI redistribution scheme, $\pi(\lambda)$ is rebated equally to all consumers, so each agent's post-transfer utility is $U_i + \pi(\lambda)$ for $i \in \{L, H\}$.

3 Welfare Evaluation with Alpha Weighting

3.1 Welfare Criteria

We evaluate social welfare under three criteria, each parameterised by $\alpha \in [0, 1]$, which governs the ethical weight placed on low-type consumers relative to high-type consumers.

$$W^{Randian}(\lambda, \alpha) = (1 - \lambda)(1 - \alpha)(V_H + UBI) \quad (8)$$

$$W^{Utilitarian}(\lambda, \alpha) = \lambda\alpha(V_L + UBI) + (1 - \lambda)(1 - \alpha)(V_H + UBI) \quad (9)$$

$$W^{Randian}(\lambda, \alpha) = \lambda\alpha(V_L + UBI) \quad (10)$$

Note that the utilitarian criterion decomposes exactly as $W^{Util} = W^{Rand} + W^{Rawls}$ for any (λ, α) . Three polar cases of α are examined:

α	Label	Interpretation
0	Randian	Full weight on high-types; $W = (1 - \lambda)(V_H + UBI)$
1/2	Neutral	Symmetric weighting of both types
1	Rawlsian	Full weight on low-types; $W = \lambda(V_L + UBI)$

3.2 Post-Redistribution Utilities

For each λ , the monopolist's total profit $\pi(\lambda)$ is redistributed as a Universal Basic Income:

$$UBI(\lambda) = \pi(\lambda)$$

Post-transfer utilities are therefore

$$\tilde{V}_H(\lambda) = V_H(\lambda) + \pi(\lambda), \quad \tilde{V}_L(\lambda) = V_L(\lambda) + \pi(\lambda)$$

where $V_H = (\theta_H - \theta_L) q_L^*$ is the high-type's information rent and $V_L = 0$ from the binding participation constraint.

3.3 Diagonal Cases

The three diagonal cases pair each welfare criterion with its natural α value: Randian with $\alpha = 0$, Utilitarian with $\alpha = \frac{1}{2}$, and Rawlsian with $\alpha = 1$. In each case welfare is computed as a function of $\lambda \in [0.05, 0.95]$ and the welfare-maximising population share λ^* is identified where it exists in the interior.

3.4 Ideology Gap

The gap between the Randian and Rawlsian criteria under a given α measures distributional tension:

$$\Delta(\lambda, \alpha) = W^{Rand}(\lambda, \alpha) - W^{Rawls}(\lambda, \alpha) = (1 - \alpha)[(1 - \lambda)\tilde{V}_H - \lambda\tilde{V}_L] \cdot \frac{1}{1 - \alpha + \alpha}$$

A large positive gap indicates that the Randian planner obtains substantially higher welfare than the Rawlsian at the same λ , reflecting disagreement about the optimal community composition. This function was a conceptual checker for me while building the system.

3.5 Expected Welfare

Integrating over a uniform distribution μ on $\lambda \in [0.05, 0.95]$, expected welfare under each criterion and α value is

$$E_\mu[W] = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} W(\lambda, \alpha)$$

The λ that maximizes and minimizes welfare is reported for each $(\alpha, \text{criterion})$ pair, providing a complete picture of how the optimal population composition depends on the planner's redistributive preferences.

4 Secant-Based Tangent Analysis on the Randian Welfare Curve

4.1 Setup

We work with the Randian welfare criterion ($\alpha = 0$), which places full weight on high-type consumers:

$$W^{\text{Rand}}(\lambda) = (1 - \lambda)(V_H(\lambda) + \pi(\lambda))$$

where $\pi(\lambda)$ is redistributed as UBI. The curve exhibits a kink at $\lambda = 1 - \theta_L/\theta_H = 0.70$, below which the low-type is excluded from the market and above which both types are served.

4.2 Tangency Condition

A line connecting two points $(\lambda_L, W(\lambda_L))$ and $(\lambda_R, W(\lambda_R))$ on the welfare curve is *tangent* to the curve at λ_R if and only if the secant slope equals the derivative of W at that point:

$$\underbrace{\frac{W(\lambda_R) - W(\lambda_L)}{\lambda_R - \lambda_L}}_{\text{secant slope}} = \underbrace{\left. \frac{dW}{d\lambda} \right|_{\lambda_R}}_{\text{tangent slope}}$$

Define the gap function:

$$g(\lambda_L, \lambda_R) = \left| \frac{W(\lambda_R) - W(\lambda_L)}{\lambda_R - \lambda_L} - \left. \frac{dW}{d\lambda} \right|_{\lambda_R} \right| \quad \lambda_R > \lambda_L$$

A pair $(\lambda_L^*, \lambda_R^*)$ with $g = 0$ is an exact tangency; in practice, we minimize g over a discrete grid.

4.3 2D Grid Search

We construct two grids:

$$\lambda_L \in [0.60, 0.99] \quad (\text{left pin, step } 0.005) \quad \lambda_R \in [0.60, 0.99] \quad (\text{right endpoint, step } 0.005)$$

For each valid pair with $\lambda_R > \lambda_L + 0.01$, we compute the secant slope and compare it to $dW/d\lambda$ evaluated numerically at λ_R . This yields the $n_L \times n_R$ gap matrix $\mathbf{G}$ with entries $G_{ij} = g(\lambda_{L,i}, \lambda_{R,j})$.

The optimal pair is:

$$(\lambda_L^*, \lambda_R^*) = \arg \min_{i,j} G_{ij}$$

4.4 Results

The best-fit tangent line is defined by its two endpoints $(\lambda_L^*, W(\lambda_L^*))$ and $(\lambda_R^*, W(\lambda_R^*))$, with slope:

$$s^* = \frac{W(\lambda_R^*) - W(\lambda_L^*)}{\lambda_R^* - \lambda_L^*} \approx \left. \frac{dW}{d\lambda} \right|_{\lambda_R^*}$$

The tangent line itself is:

$$T(\lambda) = W(\lambda_L^*) + s^*(\lambda - \lambda_L^*)$$

4.5 Economic Interpretation

The tangent point λ_R^* identifies the community composition at which the Randian planner is locally indifferent to a marginal increase in the share of low-income residents. In the problem, this means that the welfare cost of diluting the high-type population (the secant slope from λ_L^*) exactly equals the local rate of welfare change at λ_R^* . Left of λ_L^* the planner prefers fewer poor residents; right of λ_R^* the welfare loss accelerates and the planner strictly prefers a smaller low-income share.